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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Second Semester M.Tech (Structural Engineering) (CBCS) Degree Examinations
May - June 2024

22CST232 - DESIGN OF CONCRETE BRIDGES

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

- ❖ Assume all the missing data suitably as per Indian Standards.
- ❖ Use of IRC6, IRC 21, IRC 24 is permitted.

Unit – I		Marks	BT*	CO*	PO*
1.	a) Draw and explain components of a typical bridge.	10	L*2	1	2
	b) Explain classification of bridges.	10	L2	1	2
2.	a) Why box culverts are economical? Mention the reasons.	5	L2	1	2
	b) Mention the three conditions of loading deemed to be critical in box culvert construction.	5	L2	1	2
	c) Explain in detail various loading conditions given by IRC for design of bridges.	10	L2	1	2
3.	Design a box culvert having inside dimensions of 3m x 3m. This culvert is subjected to a dead load of 14000N/m ² and a live load of IRC Class AA tracked vehicles. Assume unit weight of soil to be 18000N/m ³ . The angle of repose of soil is 30°. Use M25 concrete and Fe415 steel. Road width is 7.5m. Span is 3.3m. Assume dead load, live loads and earth pressure being present, while no water pressure from inside.	20	L3	2	3
Unit – II					
4.	Determine the values of short span and long span bending moments in case of an interior panel of a T-beam bridge having following details. Dimension of the panel = 3mx3.5m, Loading IRC class A Loading pattern: Two wheels (each of 57kN) adjusted symmetrically with respect to centre of the panel.	20	L3	3	3
5.	Design the cross girder for the given data: Main girder spacing = 2.5m; Spacing between cross girder: 4m Depth of cross girder is 150mm Live load Class AA tracked vehicles. Use M25 concrete and Fe415 steel.	20	L3	3	3
6.	Determine the maximum bending moment and shear force in longitudinal girder with following data: Spacing of between main girder=2.5m, Spacing between cross girder =4m Clear width of roadway = 7.5m Span=16m, Live load: IRC Class AA tracked vehicles. Average thickness of wearing coat: 80mm Use M25 concrete and Fe415 steel.	20	L3	3	3
Unit – III					
7.	a) Mention the advantages of PSC bridges.	10	L3	5	3
	b) Explain Pre-tensioned and Post-tensioned PSC bridges.	10	L3	5	3

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8. Determine the pre-stressing force and eccentricity for a PSC girder with following data.

Clear span=9m, Clear width of road way = 7.5m

Footpaths = 600mm wide on either sides.

Thickness of wearing coat = 100mm at the centre of the road

Live load = IRC Class AA tracked vehicles.

Materials: M50 concrete and $f_{ci}=40\text{MPa}$.

$f_{ct}=20\text{MPa}$; Loss ration = 0.8; $f_{cw}=16.5\text{MPa}$

$f_{tt}=0$; $f_{cw}=0$

20 L3 5 3

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
